

Lec 9-1, Nov 3, Mon, 2025.

- Natural logarithmic function (7.2)

- Algebraic properties

- Graph and range

- $\ln(|f(x)|)$

- $\hookrightarrow \int \frac{f'(x)}{f(x)} dx$

- $\hookrightarrow$ logarithmic differentiation

- Natural exponential function (7.3)

- $\hookrightarrow$ Definition and basic properties

- $\hookrightarrow$ Monotonicity and graph

- $\hookrightarrow$ Algebraic properties

- General powers and general exponential functions (7.3)

- $\hookrightarrow$ Definition of general powers

- $\hookrightarrow$ Algebraic properties

- $\hookrightarrow e$ as a limit and its value

- $\hookrightarrow$ compound interest

- $\hookrightarrow$ Differentiation rules

Def. Define e to be the unique number in $(0, \infty)$ such that

$$\ln(e) = 1.$$

Historical Remarks about e

- Bernoulli, 1683, in the context of compound interest.
- Euler, 1731, in the context of logarithm. [The notation "e" credits to Euler.]

2. Algebraic properties.

Thm. For any $b \in \mathbb{R}_{>0}$ and $x \in \mathbb{R}_{>0}$:

1. $\ln(bx) = \ln b + \ln x$.
2. $\ln \frac{b}{x} = \ln b - \ln x$
3. $\ln \frac{1}{x} = -\ln x$
4. $\ln(x^r) = r \cdot \ln x$, for any $r \in \mathbb{Q}$ [rational number set].

Pf: Let $g(x) = \ln x$, $x > 0$.

1. Let $f(x) \triangleq \ln(bx)$ ($= g(bx)$), defined on $(0, \infty)$. Then

$$\underbrace{f'(x)} = g'(bx) \cdot b = \frac{1}{bx} \cdot b = \frac{1}{x}.$$

So $\exists$ constant C such that $f(x) = \ln x + C$, $\forall x \in (0, \infty)$.

$$\Rightarrow \ln(bx) = \ln x + C, \quad \forall x \in (0, \infty).$$

Substituting $x=1$ yields $\ln b = \ln 1 + C$,

So $C = \ln b$, i.e. $\ln(bx) = \ln x + \ln b$.

4. Let $f(x) = \ln(x^r)$, defined on $(0, \infty)$. Then

$$f'(x) = \frac{1}{x^r} \cdot r x^{r-1} = r \cdot \frac{1}{x}.$$

So $\exists$ constant C such that $f(x) = r \ln x + C$, $\forall x \in (0, \infty)$.

$$\Rightarrow \ln(x^r) = r \ln x + C.$$

Substituting $x=1$ yields $\ln 1 = r \ln 1 + C$, i.e., $C=0$.

$$\text{so } \ln(x^r) = r \ln x.$$

$$2. \ln\left(\frac{b}{x}\right) = \ln(b \cdot x^{-1}) = \ln(b) + \ln(x^{-1}) \quad [\text{by } 1]$$

$$= \ln(b) - \ln(x). \quad [\text{by } 4]$$

$$3. \ln\left(\frac{1}{x}\right) = \ln(x^{-1}) \stackrel{\text{by } 2}{=} -\ln x.$$

□.

3. Graph and range.

• Concavity

Since $\ln''(x) = \left(\frac{1}{x}\right)' = -x^{-2} < 0$ for all $x \in (0, \infty)$,

the curve $y = \ln x$ is concave down on $(0, \infty)$.

• Range

Since $\ln 2 = \int_1^2 \frac{1}{t} dt > (2-1) \cdot \frac{1}{2} = \frac{1}{2}$,

it follows that for any $n \in \mathbb{Z}_+$,

$$\ln(2^n) = n \cdot \ln 2 > \frac{1}{2} \cdot n$$

For any fixed M , by choosing n with $\frac{n}{2} \geq M$, we have $\ln(2^n) = \ln(2^n)$ $> \frac{n}{2}$

$$\ln(2^n) > \frac{n}{2} \geq M$$

$$\geq \underline{M}$$

Since $\ln$ is increasing, we see that $\ln(x)$ can exceed any fixed M

for all sufficiently large x : $\lim_{x \rightarrow \infty} \ln(x) = \infty$.

[Specifically, $\forall M > 0, \exists n = \lfloor 2M \rfloor + 1$, s.t. $x = 2^n$ and $f(x) = \ln(2^n) > M$].

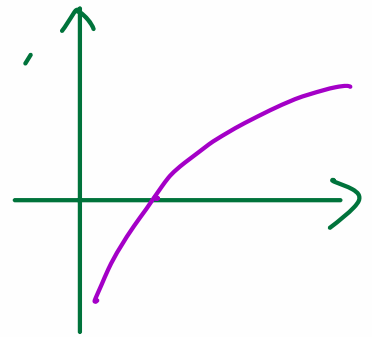
Also,

$$\lim_{x \rightarrow 0^+} \ln x \stackrel{y = \frac{1}{x}}{=} \lim_{y \rightarrow \infty} \ln\left(\frac{1}{y}\right) = \lim_{y \rightarrow \infty} (-\ln y) = -\infty;$$

So $\lim_{x \rightarrow 0^+} \ln x = -\infty$ and $x=0$ is a vertical asymptote.

Since $\lim_{x \rightarrow \infty} \ln x = \infty$ and $\lim_{x \rightarrow 0^+} \ln x = -\infty$, using IVT,

one can show that $\text{range}(\ln) = \mathbb{R} = (-\infty, \infty)$.

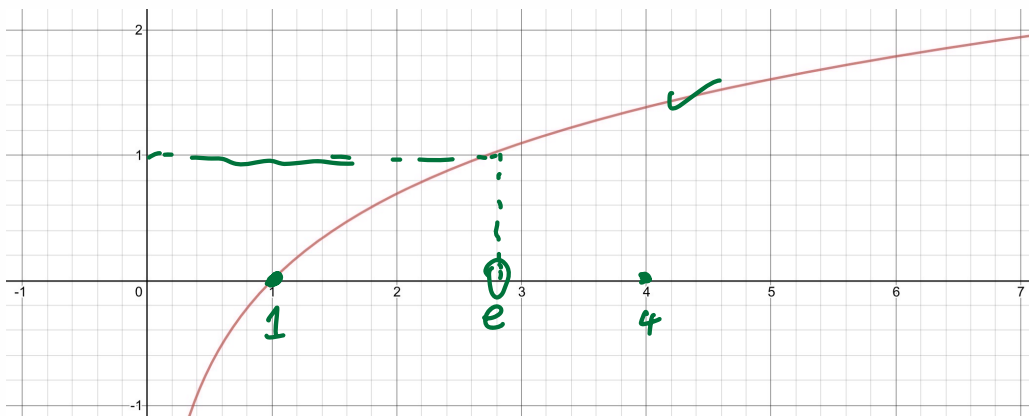


• Limits of derivatives.

Since $\lim_{x \rightarrow \infty} \ln'(x) = \lim_{x \rightarrow \infty} \frac{1}{x} = 0$, the graph $y = \ln x$ tends to

have a "flat" tangent line as x grows larger, yet it is unbounded

above.



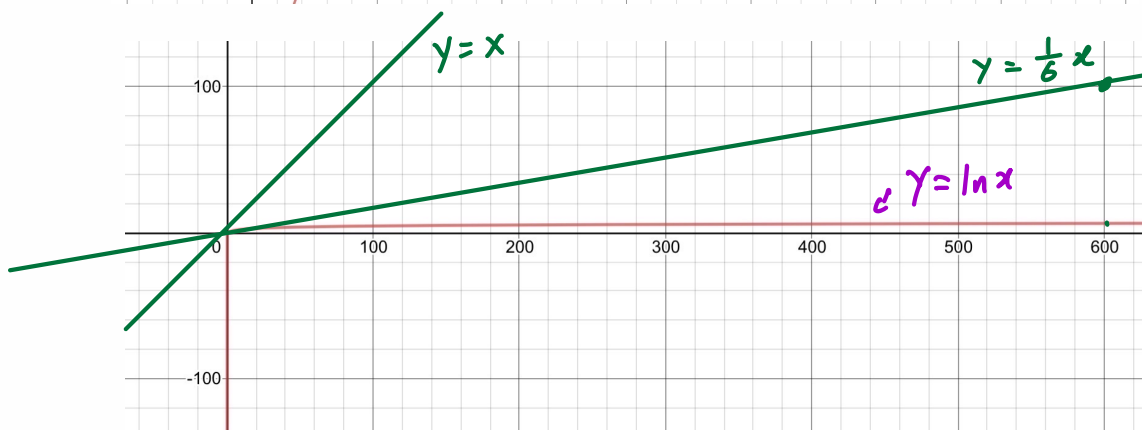
$\ln x$

① $\nearrow$, cont'
diff' $(0, \infty)$

② $\lim_{x \rightarrow 0^+} = -\infty$
 $\lim_{x \rightarrow \infty} = \infty$

③ concave down

④ $\lim_{x \rightarrow \infty} \ln'(x) = 0$



4. Composite functions: $\ln$ with $|f(x)|$.

By chain rule,

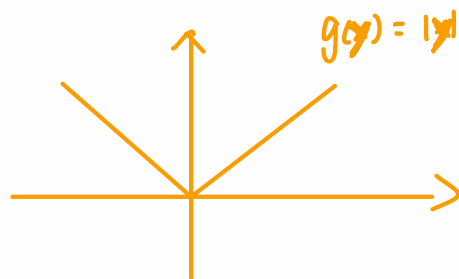
$$\frac{d}{dx} [\ln |x|] = \frac{1}{|x|} \cdot \frac{|x|}{x} = \frac{1}{x}, \quad x \in (-\infty, 0) \cup (0, \infty).$$

$$h(y) \equiv \ln(y), \quad g(x) \equiv |x| \Rightarrow \frac{d}{dx} [h(g(x))] = h'(g(x)) \cdot g'(x) = \frac{1}{g(x)} \cdot \frac{|x|}{x} = \frac{1}{x}.$$

This means that on the intervals $[-\infty, 0)$ and $(0, \infty]$,

$\ln |x|$ is an antiderivative of $\frac{1}{x}$, i.e.,

$$\int \frac{1}{x} dx = \ln |x| + C$$



holds for any one of the two intervals above.

$$[\text{Here we use } g(y) \equiv |y| \text{ and } g'(y) = \frac{|y|}{y} = \begin{cases} 1, & y > 0 \\ -1, & y < 0 \end{cases}]$$

More generally, if $h(x) = |f(x)|$ where f is differentiable and never zero,

then by the chain rule,

$$h'(x) = [g(f(x))]'$$

$$= g'(f(x)) \cdot f'(x)$$

$$= \frac{|f(x)|}{f(x)} \cdot f'(x).$$

[where $g(y) = |y|$].

So

$$\frac{d}{dx} \ln |f(x)| = \frac{d}{dx} [\ln h(x)]$$

$$= \frac{1}{h(x)} h'(x) = \frac{1}{|f(x)|} \cdot \frac{|f(x)|}{f(x)} \cdot f'(x) = \frac{f'(x)}{f(x)}.$$

That is,

$$\int \frac{f'(x)}{f(x)} dx = \ln |f(x)| + C$$

Note that it only works on an interval on which f is defined and never zero.

Extended Discussion

• The quantity $\frac{f'(x)}{f(x)}$ is called the relative rate of change of f at $x = x_0$.

• If $f(x) > 0$, $\forall x$, then $\frac{d}{dx}[\ln f(x)] = \frac{f'(x)}{f(x)}$ gives the function of relative rate of change.

= instant' rate of change of $\ln f(x)$

= relative rate of change of $f(x)$.

• Recall:

Average rate of change : $\frac{\Delta y}{\Delta x} = \frac{f(x_0 + \Delta x) - f(x_0)}{\Delta x}$

Instant rate of change = $\lim_{\Delta x \rightarrow 0} \frac{f(x_0 + \Delta x) - f(x_0)}{\Delta x} = f'(x_0)$.

e.g.1. Find $\int \tan x \, dx$ and $\int \sec x \, dx$.

$$\begin{aligned}\text{Sol: } \int \tan x \, dx &= \int \frac{\sin x}{\cos x} \, dx \\&= \int \frac{-1}{\cos x} d(\cos x) \\&\stackrel{u=\cos x}{=} - \int \frac{1}{u} \, du \Big|_{u=\cos x}\end{aligned}$$

$$= -\ln|u| \Big|_{u=\cos x} + C$$

$$= -\ln|\cos x| + C$$

Valid on intervals with $\cos$ never 0

$$\begin{aligned}& (= \ln \left| \frac{1}{\cos x} \right| + C) \quad \square \\& = \ln|\sec x| + C \quad \square\end{aligned}$$

$$\int \sec x \, dx = \int \frac{1}{\cos x} \, dx = \int \frac{\cos x}{\cos^2 x} \, dx$$

$$= \int \frac{1}{1-\sin^2 x} d(\sin x)$$

$$\stackrel{u=\sin x}{=} \int \frac{1}{1-u^2} \, du \Big|_{u=\sin x} = \frac{1}{2} \int \frac{1}{1-u} + \frac{1}{1+u} \, du \Big|_{u=\sin x}$$

$$= \frac{1}{2} \int \frac{1}{1+u} \, du \Big|_{u=\sin x} - \frac{1}{2} \int \frac{1}{u-1} \, du \Big|_{u=\sin x}$$

$$= \frac{1}{2} \ln|1+u| \Big|_{u=\sin x} - \frac{1}{2} \ln|u-1| \Big|_{u=\sin x} + C$$

$$= \frac{1}{2} \ln|1+\sin x| - \frac{1}{2} \ln|\sin x-1| + C$$

$$= \frac{1}{2} [\ln|1+\sin x| - \ln|\sin x-1|] + C$$

$\square$

Valid on intervals with

$$\begin{aligned}\tan x &= \frac{\sin x}{\cos x} \\ \sin^2 x + \cos^2 x &= 1\end{aligned}$$

(or: $\frac{1}{2} \ln \left| \frac{1+\sin x}{\sin x - 1} \right| + C = \frac{1}{2} \ln \left| \frac{\cos^2 \frac{x}{2} + \sin^2 \frac{x}{2} + 2\sin \frac{x}{2} \cos \frac{x}{2}}{2\sin \frac{x}{2} \cos \frac{x}{2} - \cos^2 \frac{x}{2} - \sin^2 \frac{x}{2}} \right| + C$ sin never ± 1 .

$$= \frac{1}{2} \ln \left| \frac{(\cos \frac{x}{2} + \sin \frac{x}{2})^2}{(\cos \frac{x}{2} - \sin \frac{x}{2})^2} \right| + C$$

$$= \ln \left| \frac{\cos \frac{x}{2} + \sin \frac{x}{2}}{\cos \frac{x}{2} - \sin \frac{x}{2}} \right| + C$$

/

$$= \ln \left| \frac{(\cos \frac{x}{2} + \sin \frac{x}{2})(\cos \frac{x}{2} + \sin \frac{x}{2})}{(\cos \frac{x}{2} - \sin \frac{x}{2})(\cos \frac{x}{2} + \sin \frac{x}{2})} \right| + C$$

$$= \ln \left| \frac{\sin x + 1}{\cos x} \right| + C$$

$$= \ln | \sec x + \tan x | + C . \quad \checkmark \quad) \quad \square$$

↖ valid on interval with
 $\sec x + \tan x \neq 0$,

i.e. $\sin x \neq \pm 1$.

Formulae:

$$\int \tan x \, dx = -\ln |\cos x| + C = \ln |\sec x| + C.$$

$$\int \sec x \, dx = \ln |\sec x + \tan x| + C.$$

e.g. 2 $\int_0^{\pi/6} \tan(2x) \, dx = ?$

Sol: LHS $\overset{\substack{z=2x \\ x=\frac{z}{2}}}{\int_0^{\pi/6}} \tan z \, d(\frac{z}{2})$

$$= \frac{1}{2} \int_0^{\pi/3} \tan z \, dz$$

$$= \frac{1}{2} \ln |\sec z| \Big|_0^{\pi/3}$$

$$= \frac{1}{2} \left(\ln \left| \frac{1}{\cos \pi/3} \right| - \ln \left| \frac{1}{\cos 0} \right| \right)$$

$$= \frac{1}{2} \left(\ln \left| \frac{1}{\frac{1}{2}} \right| - \ln |1| \right)$$

$$= \frac{1}{2} \ln 2. \quad \square$$

$$\text{LHS} = \frac{1}{2} \int_0^{\pi/3} \tan z \, dz$$

$$= \frac{1}{2} \int_0^{\pi/3} \frac{\sin z}{\cos z} \, dz = -\frac{1}{2} \int_0^{\pi/3} \frac{1}{\cos z} \, d \cos z$$

$$\overset{u=\cos z}{=} -\frac{1}{2} \int_{\cos 0}^{\cos \pi/3} \frac{1}{u} \, du$$

$$= -\frac{1}{2} \left(\ln \left| \frac{1}{2} \right| - \ln |1| \right) = \frac{1}{2} \ln 2. \quad \square$$

Logarithmic differentiation

e.g.3. Find y' for $x > 0$, where $y = \frac{x^{3/4} (x^2+1)^{1/2}}{(3x+2)^5}$.

Sol: [complicated to directly take a derivative]

[Idea: Take $\ln$!]

$$\because x > 0 \quad \therefore$$

$$\therefore \ln y = \frac{3}{4} \ln x + \frac{1}{2} \ln(x^2+1) - 5 \ln(3x+2).$$

$\therefore$ Applying $\frac{d}{dx}$ to both sides:

$$\frac{d}{dx}(\ln y) = \frac{1}{y} \cdot y' = \frac{3}{4} \frac{1}{x} + \frac{1}{2} \frac{1}{x^2+1} \cdot 2x - 5 \cdot \frac{1}{3x+2} \cdot 3$$

$$\therefore y' = y \cdot \left[\frac{3}{4} \frac{1}{x} + \frac{1}{2} \frac{1}{x^2+1} \cdot 2x - 5 \cdot \frac{1}{3x+2} \cdot 3 \right]$$

$$= \frac{x^{3/4} (x^2+1)^{1/2}}{(3x+2)^5} \cdot \left[\frac{3}{4} \frac{1}{x} + \frac{1}{2} \frac{1}{x^2+1} \cdot 2x - 5 \frac{1}{3x+2} \cdot 3 \right]$$

□.

This approach of taking $\ln$ may be used to differentiate $F(x)$ with the form

$$F(x) = \prod_{k=1}^n (f_k(x))^{m_k} \triangleq f_1(x)^{m_1} f_2(x)^{m_2} \cdots f_n(x)^{m_n},$$

↑ " $\prod$ " = " P_i " = "Product".

where $\underline{f_k(x)} > 0$ and m_k may be negative.

$$\therefore y = F(x).$$

$$\hookrightarrow \ln y = m_1 \ln f_1(x) + m_2 \ln f_2(x) + \dots + m_n \ln f_n(x).$$

$$\hookrightarrow \frac{y'}{y} = m_1 \frac{f_1'(x)}{f_1(x)} + m_2 \cdot \frac{1}{f_2(x)} f_2'(x) + \dots + m_n \cdot \frac{f_n'(x)}{f_n(x)}.$$

$$\hookrightarrow y' = y \cdot \left[m_1 \frac{f_1'(x)}{f_1(x)} + m_2 \frac{f_2'(x)}{f_2(x)} + \dots + m_n \frac{f_n'(x)}{f_n(x)} \right].$$

If $f_k(x) < 0$ for some k above, we may replace $F(x)$ with $|F(x)|$ when working with $\ln$:

$$\frac{d}{dx} \ln |F(x)| = \frac{F'(x)}{F(x)} \Rightarrow F'(x) = F(x) \cdot \frac{d}{dx} \ln |F(x)|.$$

e.g. 4. Let $y = \frac{(2x+1)^3}{(3x-1)^5}$. Find y' using logarithmic differentiation.

$$\text{Sol: } |y| = \frac{|2x+1|^3}{|3x-1|^5}.$$

$$\ln |y| = 3 \ln |2x+1| - 5 \ln |3x-1|.$$

$$\frac{1}{|y|} \cdot \frac{|y|}{y} \cdot y' = 3 \cdot \frac{1}{|2x+1|} \cdot \frac{|2x+1|}{2x+1} \cdot 2 - 5 \cdot \frac{1}{|3x-1|} \cdot 3$$

$$\frac{y'}{y} = 3 \cdot \frac{2}{2x+1} - 5 \cdot \frac{3}{3x-1}.$$

$$y' = \frac{(2x+1)^3}{(3x-1)^5} \left(3 \cdot \frac{2}{2x+1} - 5 \cdot \frac{3}{3x-1} \right).$$

□

• Natural exponential function.

1. Definition and basic properties.

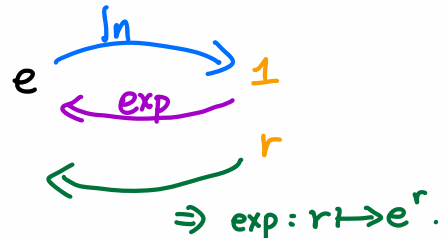
one-to-one

- Since $\ln$ is injective on $(0, \infty)$ and has the range $\mathbb{R}$, it has an inverse function.

Def: The (natural) exponential function is the inverse function of $\ln$, denoted by $\exp: \mathbb{R} \rightarrow (0, \infty)$.

- Since $\ln(e) = 1$, we have

$$\exp(1) = e$$



- Since $e > 0$, e^r is defined for all rational number r .

- Since $\ln(?) = r$, we have

$e^r \xrightarrow{\ln} r$
 $r \xrightarrow{\exp} e^r$
 $\exp(r) = e^r$

$\hookrightarrow$

$$\exp(r) = e^r, \quad \forall r \in \mathbb{Q}.$$

[This is a property from definitions of $\exp$ and e .]

$\pi, 2\pi, \dots$

- For irrational power x , we have **NOT** defined e^x formally yet.

We define it as follows:

Def: $e^x \triangleq \exp(x), \quad \forall x \in \mathbb{R} \setminus \mathbb{Q}.$

[This is a definition.]

- Hence, $\exp(x) = e^x$ for any $x \in \mathbb{R}$. Consequently,

$$e^{\ln x} \stackrel{\text{def}}{=} \exp(\ln x) = x, \quad \forall x \in (0, \infty);$$

$$\ln(e^x) \stackrel{\text{def}}{=} \ln(\exp(x)) = x, \quad \forall x \in \mathbb{R}.$$

- Let $y = \exp(x)$. For any fixed $x_0 \in \mathbb{R}$, let $y_0 = \exp(x_0)$. Then
Let $g = \ln$.

$$(g^{-1})'(x_0) = \exp'(x_0) \stackrel[\text{diff.}]{\text{inverse}} \left. \frac{1}{(g)'(y_0)} \right|_{y_0 = g^{-1}(x_0)}$$

$$= \left. \frac{1}{(\ln)'(y_0)} \right|_{y_0 = \exp(x_0)}$$

$$= \left. \frac{1}{\frac{1}{y_0}} \right|_{y_0 = \exp(x_0)}$$

$$= \exp(x_0) \quad \text{Q.E.D.}$$

$\therefore \exp'(x) = \exp(x), \quad \text{or} \quad \frac{d}{dx} e^x = e^x, \quad \forall x \in \mathbb{R}.$

2. Monotonicity and graph

• Since $\exp'(x) = \exp''(x) = \exp(x) > 0$, $\forall x \in \mathbb{R}$,
the function $\exp$ is **increasing** and **concave up** on $\mathbb{R}$.

• Using limit definitions, one can prove $\lim_{x \rightarrow -\infty} e^x = 0$

$$\text{and } \lim_{x \rightarrow \infty} e^x = \infty.$$

[Proof: optional exercise.]

3. Algebraic properties

Thm [Algebraic properties of $\exp$] For any real numbers x_1, x_2 and x :

$$1. e^{x_1} e^{x_2} = e^{x_1 + x_2}.$$

$$2. e^{-x} = \frac{1}{e^x}$$

$$3. e^{x_1} / e^{x_2} = e^{x_1 - x_2}.$$

$$4. (e^x)^r = e^{rx}, \quad \forall r \in \mathbb{Q}.$$

Pf: 1. Let $y_i = e^{x_i}$ for $i \in \{1, 2\}$. Then

$$\underline{e^{x_1} e^{x_2}} = y_1 y_2 = e^{\ln(y_1 y_2)} = e^{\ln y_1 + \ln y_2} = \underline{e^{x_1 + x_2}} \quad \square$$

$\uparrow$
ln property 1

4. Let $y = e^x$. Then

$$(e^x)^r = y^r = e^{\ln y^r} = e^{r \ln y} = e^{rx} \quad \square$$

$\uparrow$
ln property 4

$$\underline{3.} \quad e^{x_1 - x_2} = e^{x_1} e^{-x_2} = e^{x_1} (e^{x_2})^{-1} = e^{x_1} / e^{x_2} \quad \square$$

$\uparrow$ exp property 1 $\uparrow$ exp property 4

$$\underline{2.} \quad e^{-x} = (e^x)^{-1} = \frac{1}{e^x} \quad \square$$

$\uparrow$
exp property 4

$$[\text{or: } e^{-x} = e^{0-x} = e^0 / e^x = 1/e^x.]$$

$\uparrow$
exp property 3

General powers !

With e^x defined for all $x \in \mathbb{R}$, we can now define a^x for any $a \in (0, \infty)$ and $x \in \mathbb{R}$.

Def. For any $a \in (0, \infty)$ and $x \in \mathbb{R}$, define

$$a^x \triangleq e^{x \ln a}.$$

Algebraic property 4 for $\ln$ and $\exp$ now also holds for irrational values of r . That is,

$$(i) \ln(x^a) = a \ln x, \quad \forall x \in (0, \infty), \quad \forall a \in \mathbb{R}.$$

$$(ii) (e^x)^a = e^{ax}, \quad \forall x \in \mathbb{R}, \quad \forall a \in \mathbb{R}.$$

Both can be verified using algebraic properties / definitions:

(i) holds because

$$\ln(x^a) \stackrel{\text{def}}{=} \ln(e^{a \ln x}) = a \ln x.$$

(ii) holds by using (i). [Details : exercise].

$$(e^x)^a \stackrel{\text{def}}{=} e^{a \ln e^x} = e^{ax}. \quad \square$$

1. Algebraic properties

The algebraic properties for e^x also hold for a^x .

Thm. For any $a \in (0, \infty)$ and any real numbers x, x_1, x_2 and r :

$$1. a^{x_1} a^{x_2} = a^{x_1 + x_2}.$$

$$2. a^{-x} = \frac{1}{a^x}$$

$$3. a^{x_1} / a^{x_2} = a^{x_1 - x_2}.$$

$$4. (a^x)^r = a^{rx}.$$

The proofs follow from definitions and properties of $\ln$ and $\exp$.

$$\begin{aligned} \text{Pf: } 1. \quad a^{x_1} a^{x_2} &\stackrel{\text{def}}{=} e^{x_1 \ln a} \cdot e^{x_2 \ln a} \\ &\stackrel{\text{exp property}}{=} e^{x_1 \ln a + x_2 \ln a} \\ &= e^{(x_1 + x_2) \ln a} \stackrel{\text{def of } a^x}{=} a^{x_1 + x_2} \quad \square \end{aligned}$$

2, 3, 4: exercise. □

2. The value of e

So far, we see that $1 < e < 4$.

[Recall: e is defined to be the unique number such that $\ln e = 1$.]

To see a more accurate approximation, we are going to show:

Thm: $e = \lim_{x \rightarrow 0} (1+x)^{\frac{1}{x}}$

$$x=1 \Rightarrow (1+1)^1 = 2$$

$$x=0.5 \Rightarrow (1+\frac{1}{2})^2 = 2.25$$

$$x=0.1 \Rightarrow (1+0.1)^{10} \approx \dots$$

By looking at values of $(1+x)^{\frac{1}{x}}$ with x getting close to 0, we get

$$e \approx 2.71828 \dots$$

Pf of Thm:

$$(1+x)^{\frac{1}{x}} \stackrel{\text{def}}{=} e^{\frac{1}{x} \ln(1+x)}$$

$$\lim_{x \rightarrow 0} \frac{1}{x} \ln(1+x) = \lim_{x \rightarrow 0} \frac{1}{x} \int_1^{1+x} \frac{1}{t} dt$$

$$= \lim_{x \rightarrow 0} \frac{F(x) - F(0)}{x}$$

$$= F'(0)$$

$$\stackrel{\text{FTC1}}{=} \left. \frac{1}{1+x} \right|_{x=0}$$

$$= 1.$$

Write

$$F(x) = \int_1^{1+x} \frac{1}{t} dt.$$

$$\Rightarrow F'(x) \stackrel{\text{FTC1}}{=} \frac{1}{1+x}$$

$\Rightarrow$

• Since $\exp$ is cont' ,

$$\lim_{x \rightarrow 0} (1+x)^{1/x} = \lim_{x \rightarrow 0} e^{\frac{1}{x} \ln(1+x)}$$

$$= e^{\lim_{x \rightarrow 0} \frac{1}{x} \ln(1+x)}$$

$$= e^1 = e.$$

□.

By letting $y = \frac{1}{x}$, we have

$$\lim_{x \rightarrow 0^+} (1+x)^{1/x} = \lim_{y \rightarrow \infty} \left(1 + \frac{1}{y}\right)^y.$$

As $x \rightarrow 0^+$, we have $y \rightarrow \infty$, so

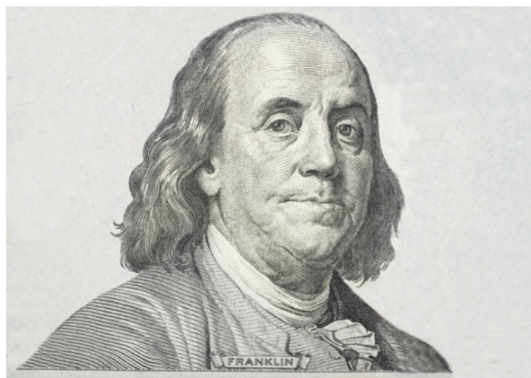
$$e = \lim_{x \rightarrow 0^+} (1+x)^{1/x} = \lim_{y \rightarrow \infty} \left(1 + \frac{1}{y}\right)^y$$

This gives an alternative limit form of e :

$$e = \lim_{y \rightarrow \infty} \left(1 + \frac{1}{y}\right)^y.$$

Compound interests and Euler's number

"Money makes money. And the money that money makes, makes money." — Benjamin Franklin.



Suppose you have \$1, and you put it in a bank with annual interest 100%. Suppose the interest is compounded n times a year. Then at the end of the year, you will get:

- ▶ $1 + 1 = 2$ dollars if $n = 1$.
- ▶ $(1 + \frac{1}{2})^2 = 2.25$ dollars if $n = 2$.
- ▶ $(1 + \frac{1}{4})^4 = 2.4414$ dollars if $n = 4$.
- ▶ $(1 + \frac{1}{n})^n$ dollars in general.

• For $n=1$, total money = $1 + (\text{interest}) 1 \cdot 100\% = \2 .

• For $n=2$, at the half of the year, you get $1 + (\text{interest}) 1 \cdot 100\% \cdot \frac{1}{2} = \1.5 .

at the second half . you get $1.5 + (\text{interest}) 1.5 \cdot 100\% \cdot \frac{1}{2} = 1.5^2 = \2.25 .

• For $n=4$, at the 1st quarter, you get $1 + \text{interest } \frac{1}{4} \cdot 100\% \cdot 1 = 1.25 \$$

2nd , $1.25 + \text{interest } \frac{1}{4} \cdot 100\% \cdot 1.25 = 1.25^2 \$$

3rd , $1.25^2 + \text{interest } \frac{1}{4} \cdot 100\% \cdot 1.25^2 = 1.25^3 \$$

4th , $1.25^3 + \text{interest } \frac{1}{4} \cdot 100\% \cdot 1.25^3 = 1.25^4 \$$.

• For n in general, you get $(1 + 100\% \cdot \frac{1}{n} \cdot 1)^n = (1 + \frac{1}{n})^n$ \$.

• When n is very large, you expect to get

$$\lim_{n \rightarrow \infty} (1 + \frac{1}{n})^n = e \approx 2.718 \dots$$

3. Differentiation rules

With general powers, we can define two types of functions:

1. (General exponential function, with base a)

Fix $a \in (0, \infty)$. Define $f(x) = a^x$, $D = \mathbb{R}$.

2. (General power function, with power a)

Fix $a \in \mathbb{R}$. Define $f(x) = x^a$, $D = (0, \infty)$.

Recall that for power functions, we have been using the rule

$$\frac{d}{dx} x^a = a x^{a-1} \quad \text{for } x \in (0, \infty).$$

$$\frac{d}{dx} x^r = r x^{r-1}.$$

We have only proved it for $r \in \mathbb{Q}$. Now we can prove it generally.

$$\text{Pf: } \frac{d}{dx} x^a \stackrel{\text{def}}{=} \frac{d}{dx} [e^{a \ln x}]$$

$$= e^{a \ln x} \cdot \frac{a}{x}$$

$$= x^a \cdot \frac{a}{x}$$

$$= a \cdot x^{a-1}, \quad \forall a \in \mathbb{R}.$$

□.

Similar techniques can be used to find $\frac{d}{dx} x^x$ (for $x > 0$):

$$\text{Sol: } \frac{d}{dx} (x^x) = \frac{d}{dx} (e^{x \ln x})$$

$$= e^{x \ln x} [\ln x + x \cdot \frac{1}{x}]$$

$$= x^x [\ln x + 1].$$

□.

□.

For a fixed $a \in (0, \infty)$, we have

$$\frac{d}{dx} (a^x) = \frac{d}{dx} (e^{x \ln a})$$

$$= e^{x \ln a} \cdot \ln a$$

$$= a^x \cdot \ln a$$

□

$\Rightarrow$

$$\boxed{\frac{d}{dx} a^x = a^x \cdot \ln a.}$$

In other words, provided that $a \neq 1$,

$$\int a^x dx = \frac{a^x}{\ln a} + C \quad \text{on } \mathbb{R}.$$

• Graph of $f(x) = a^x$ ($a \neq 1$)

$$\textcircled{1} f'(x) = \begin{cases} a^x \ln a > 0 & , \text{ if } a > 1 \\ a^x \ln a < 0 & , \text{ if } 0 < a < 1 \end{cases} \quad \text{for any } x.$$

$$\Rightarrow f \text{ is } \begin{cases} \text{increasing} & , \text{ if } a > 1 \\ \text{decreasing} & , \text{ if } a \in (0, 1) \end{cases} \quad \text{for any } x.$$

$$\textcircled{2} f''(x) = a^x \ln a \cdot \ln a > 0, \quad \forall x$$

$$\Rightarrow f \text{ is concave up on } \mathbb{R}.$$

$\textcircled{3}$ limits :

$$\lim_{x \rightarrow \infty} a^x = ? \quad \left(\begin{array}{l} \infty, \text{ if } a > 1; \\ 0, \text{ if } a \in (0, 1) \end{array} \right)$$

$$\lim_{x \rightarrow -\infty} a^x = ? \quad \left(\begin{array}{l} 0, \text{ if } a > 1; \\ \infty, \text{ if } a \in (0, 1) \end{array} \right)$$

Proof : exercise.

